

Blister Internship Summer 2023 Report and Findings

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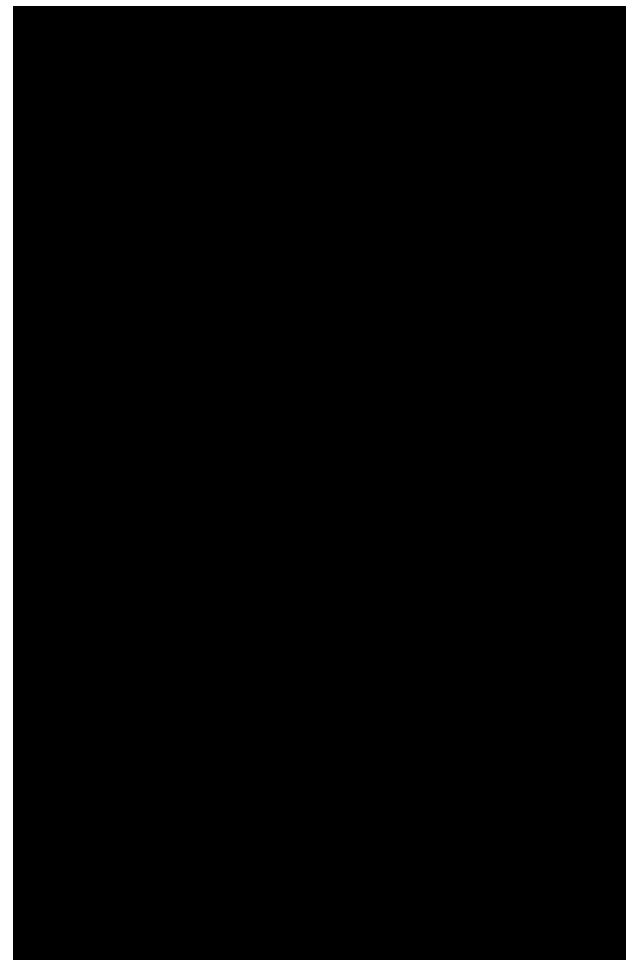
Mechanical Engineering
University of Colorado Boulder

Summary of Results

- Finished the development of 4 laboratory tests that quantify radial and lateral characteristics of mountain bike wheelsets
- All tests this Summer were conducted on wheelsets specifically categorized as all mountain/enduro
- Carbon vs Aluminum is not as simple as it seems
- Field tests have taken a step forward but are still not capable of accurately measuring target values in an uncontrolled environment
- Have devised a questionnaire and rider/wheel categorization based on a mathematical model; influenced by lab tests, rider input, and user experience

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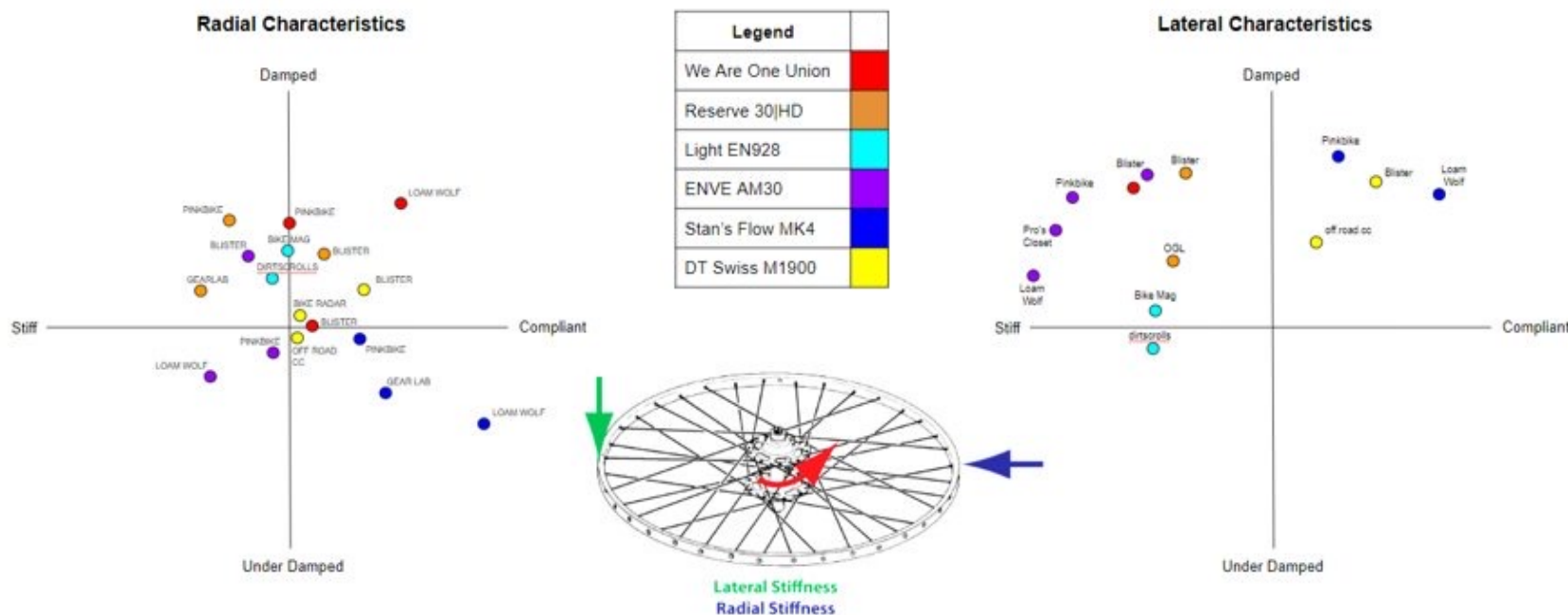
1. Qualitative Online Reviews
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7. Rider Questionnaire and Regression Model



Qualitative Online Reviews

Online Review Survey

- Combed through online reviews about the wheels on test and contacted David Golay to verify
- Plotted qualitative results on the chart below and defined common verbiage used in reviews



Vocab List (Qualitative → Quantitative)

Stiff: Wheel has low deformation under load

Compliant: Wheel has high deformation under load

Damped: Harmonic oscillations reduce in amplitude rapidly; damping ratio ≈ 1 , short settling time. The wheel absorbs and dissipates energy more effectively through rapid impacts

Underdamped: Harmonic oscillations continue without dissipating effectively; damping ratio < 1 . The wheel cannot absorb and dissipate energy through rapid impacts

Chatter: Occurs when a wheel is underdamped and radially stiff, producing undesirable vibrations through rapid impacts, reducing traction.

Lively: A largely qualitative description of a wheel's ability to respond quickly to impacts and dampen chatter while maintaining radial stiffness, giving the rider feedback from the trail.

Ride Quality: The wheel's overall feel and ability to absorb vibrations/impacts from the trail. Good ride quality can vary based on rider weight; a wheel that is compliant for a 200 lb rider may be stiff for a 100 lb rider. Current industry goals for good ride quality include... Increasing lateral stiffness, increasing radial compliance, and increasing damping (reducing chatter).

Radial Quasi-Static Test

Radial Quasi-Static Test

- Goals
- Issues and Outcomes
- Testing Procedure
- Results
- Next Steps



Radial Quasi-Static Test - Goals

- Determine stress thresholds in mountain bike wheelsets
- Using the MTS and OptiTrack systems symbiotically will result in the collection of data relevant to radial stiffness.
- Simulate a low strain rate scenario such as a big g-out, jump landing/takeoff, or big berm
 - This will equate most directly to rider weight and flow trail environment

Radial Quasi-Static Test – Issues and Outcomes

- OptiTrack has a hard time tracking markers through time
 - Camera placement not ideal with where the MTS is placed
 - Calibrated system to below 0.2mm error to minimize losing markers
 - Placed markers in places where at least 3 cameras but ideally 4 can see it throughout the entire test
- Matlab code was not working well and producing graphs that were difficult to read or make conclusions based off
 - Switched to python code for easy data importing and easy to read graphs

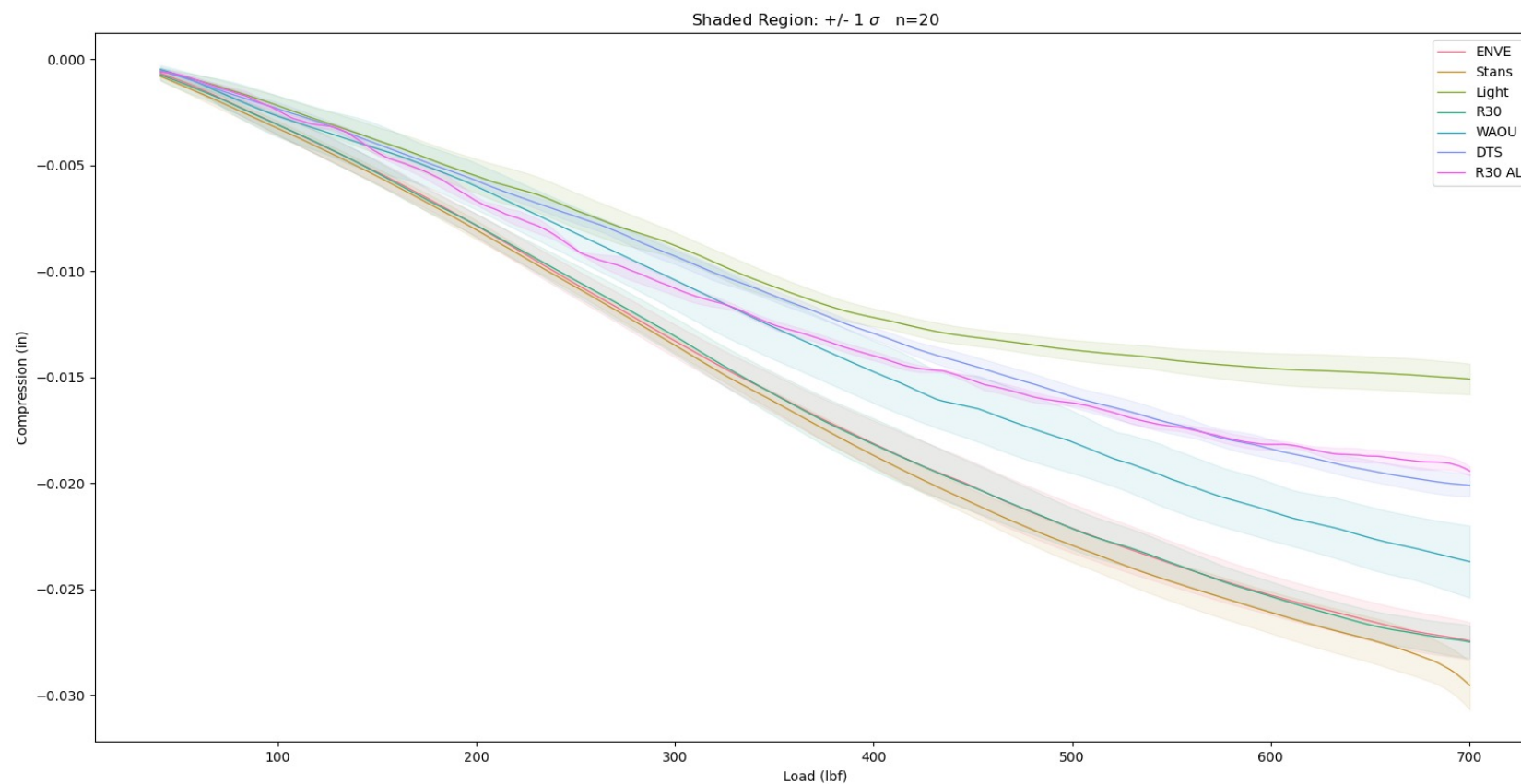
Radial Quasi-Static Test – Testing Procedure

- Follow Standard Operating Procedure
 - [Standard Operating Procedure](#)
- Wheel is trued to manufacturer spec prior to each round of testing
- Ensure that tires are inflated to 32.0 psi +/- 0.1 prior to testing
- Ensure that flat circular head is mounted to MTS
- Export data as a .csv file in order to plot results

Radial Quasi-Static Test – Results

- Tested each wheel for 25 trials
- Found wheel with lowest number of viable trials
- Randomly removed data from other wheels to match number of trails between wheels
- Reduced the noise and found the average and standard deviation

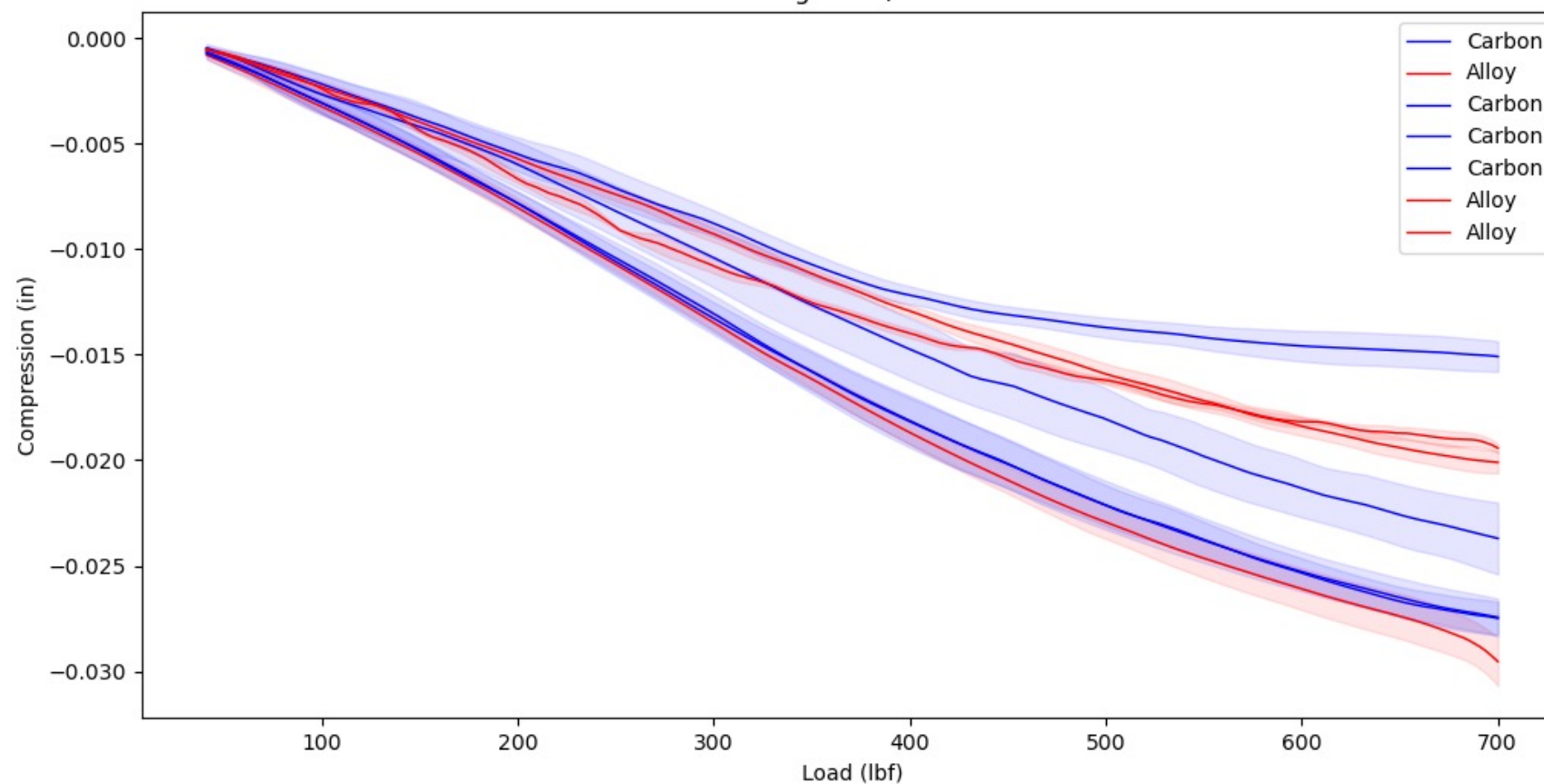
Quasi-Static Loading on Bike Rims with Tires



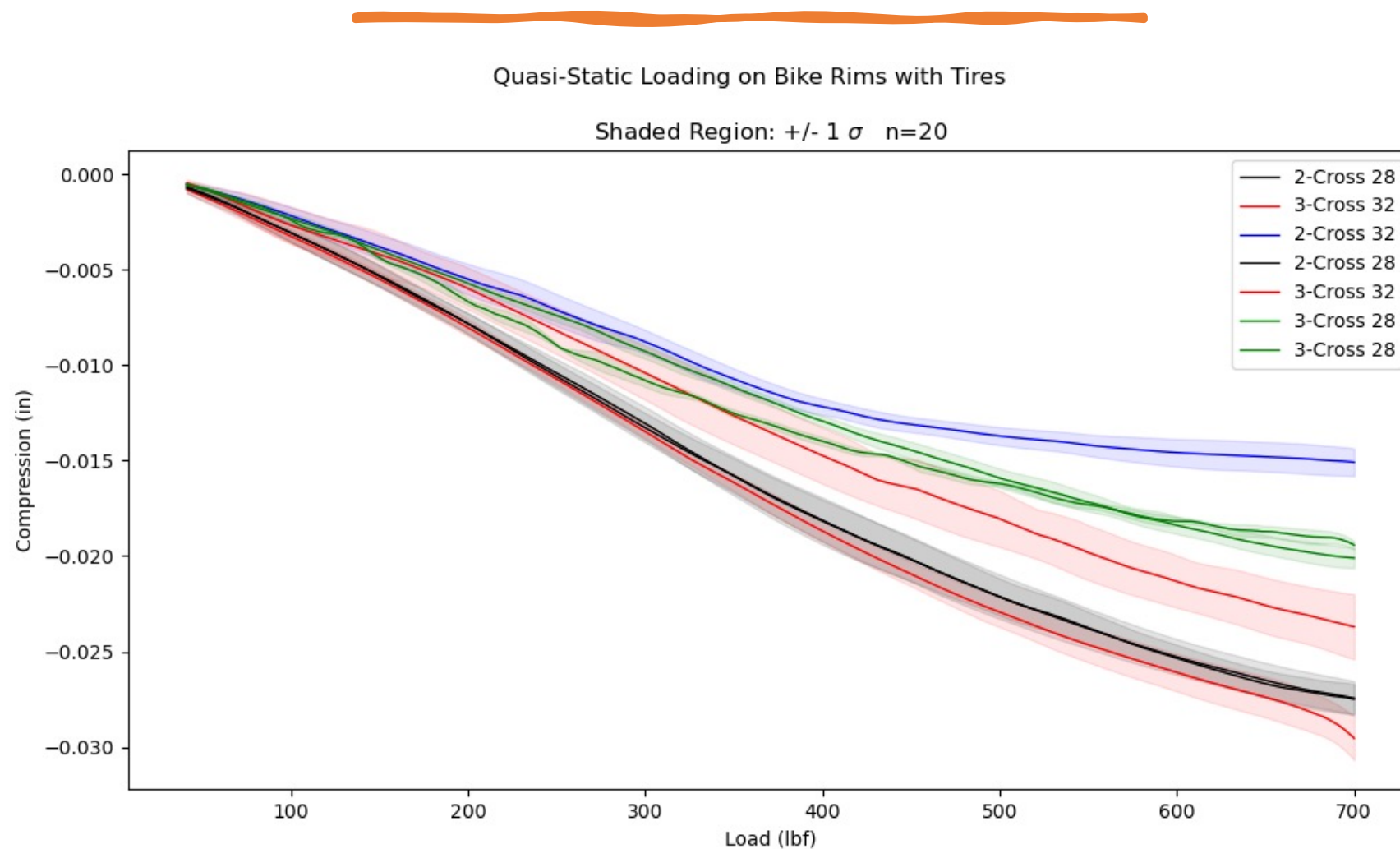
Radial Quasi-Static Test – Results

Quasi-Static Loading on Bike Rims with Tires

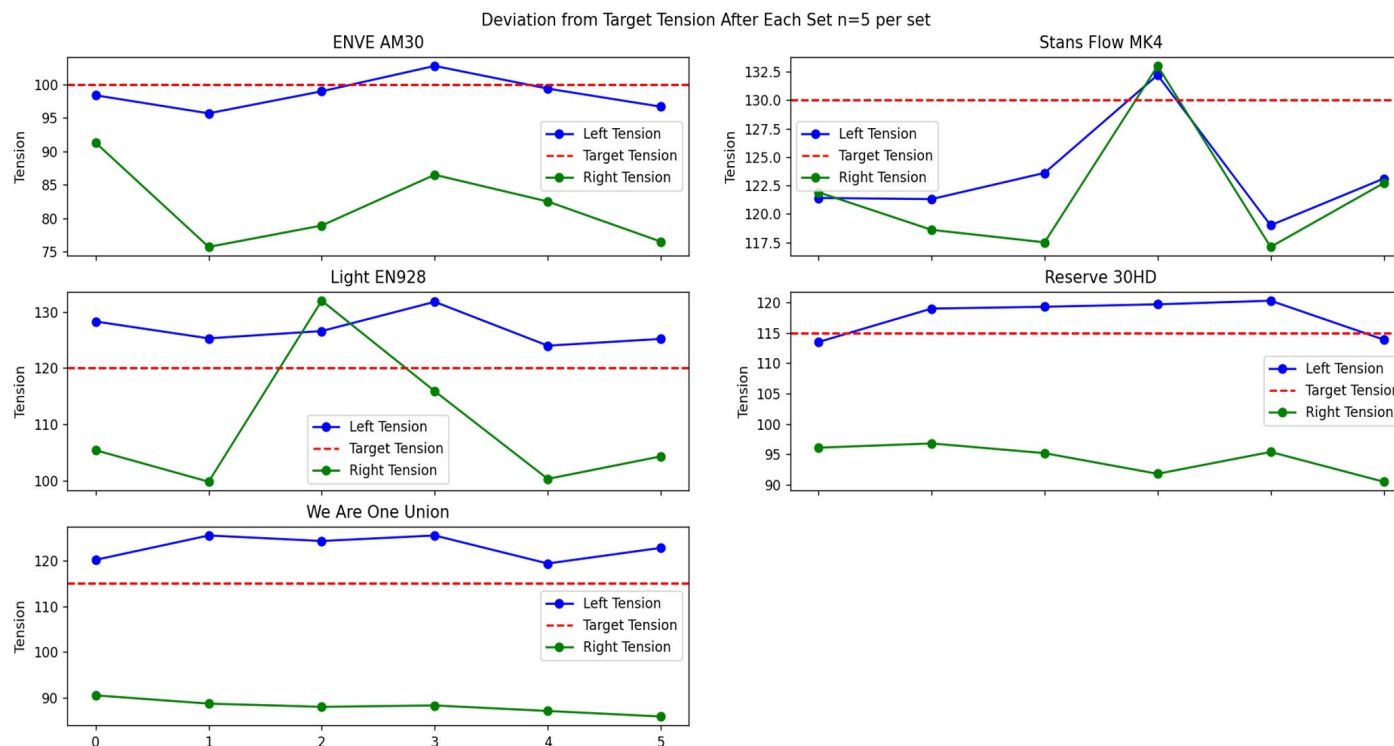
Shaded Region: $\pm 1 \sigma$ $n=20$



Radial Quasi-Static Test – Results



Radial Quasi- Static Test – Results



- Concerned with wheel trueness throughout testing so we recorded to tension and lateral deviation after each round of 5 tests
- Found no deviation throughout testing showing we are putting the wheels through normal scenarios and only need to re-true wheels before a testing session

Radial Quasi-Static Test – Next Steps

- Get more wheels on test to start investigating trends
 - Carbon vs. Aluminum
 - Spoke Count
 - Spoke Diameter
 - Spoke Cross Pattern (Lacing)
 - Rim Profile

Radial Dynamic Load Test

Radial Dynamic Load Test

- Goals
- Issues and Outcomes
- Testing Procedure
- Results
- Next Steps



Radial Dynamic Load Test - Goals

- The objective of the dynamic testing procedure is to determine energy dissipation, trends in deformation, and harmonic oscillations of the wheels as a result of impacts on mountain bike wheelsets.
- This testing is in efforts to better understand the wheel characteristics
- Understanding how the wheel systems' harmonics, energy dissipation, and deflection are affected under different loading scenarios in hopes of matching riders with their "perfect" wheel system.

Radial Dynamic Load Test - Issues and Outcomes

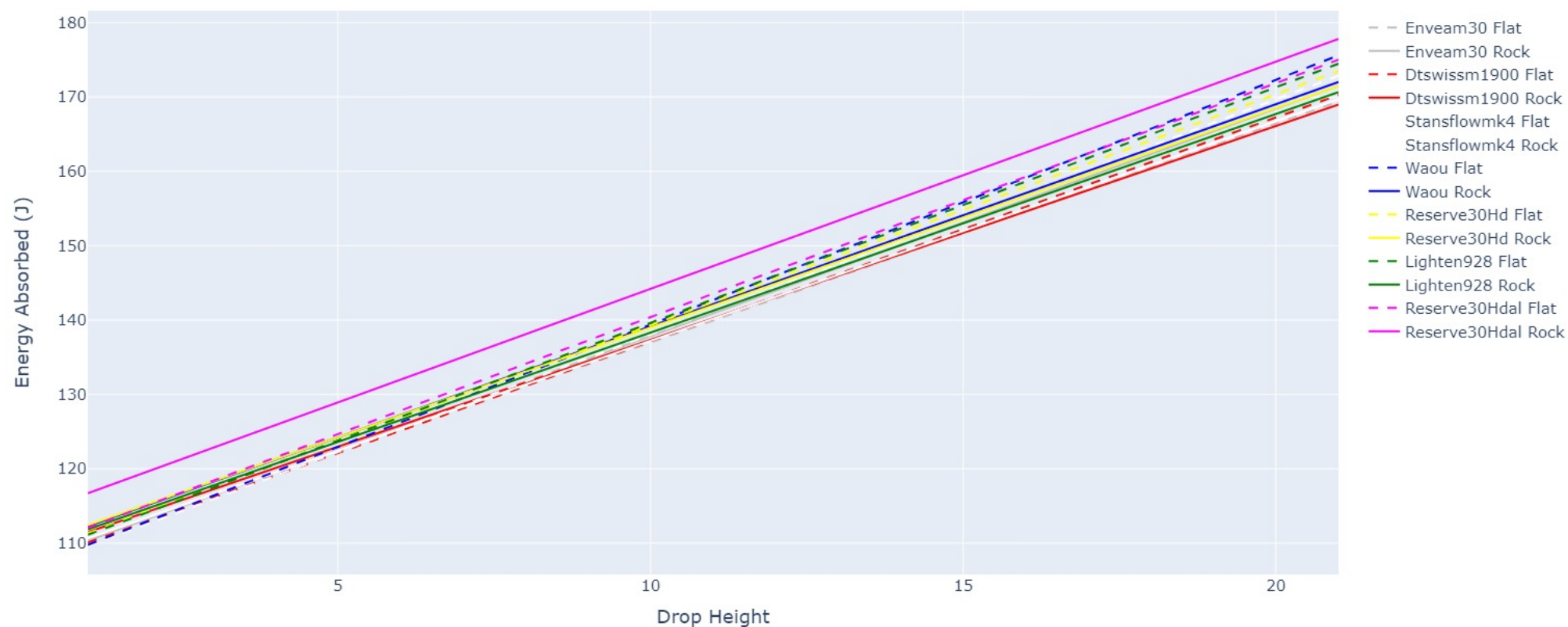
- Rigidity.
 - Due the flexing and vibration that occurs in the testing fixture, we are unable to efficiently capture the vibrations associated with the impact of the wheelsets using the OptiTrack System
- Design Oversights.
 - The testing sled and impactor, which is like that of a cantilever beam, applies a torsional load to the roller bearings as well as the wheelset which can interfere with the collection of data.
 - Would need to modify the entire
- Automation.
 - If a multitude of tests are being performed, then having automation of some sort can prove to be very beneficial as an alternative to carrying the carriage each test.

Radial Dynamic Load Test - Testing Procedure

- Follow Standard Operating Procedure
 - [Standard Operating Procedure](#)
- Wheel is trued to manufacturer spec prior to each round of testing
- Ensure that tires are inflated to 32.0 psi +/- 0.1 prior to testing
- Ensure that either the flat or curved impactor is attached to the sled
- Export data as a .csv file in order to plot results

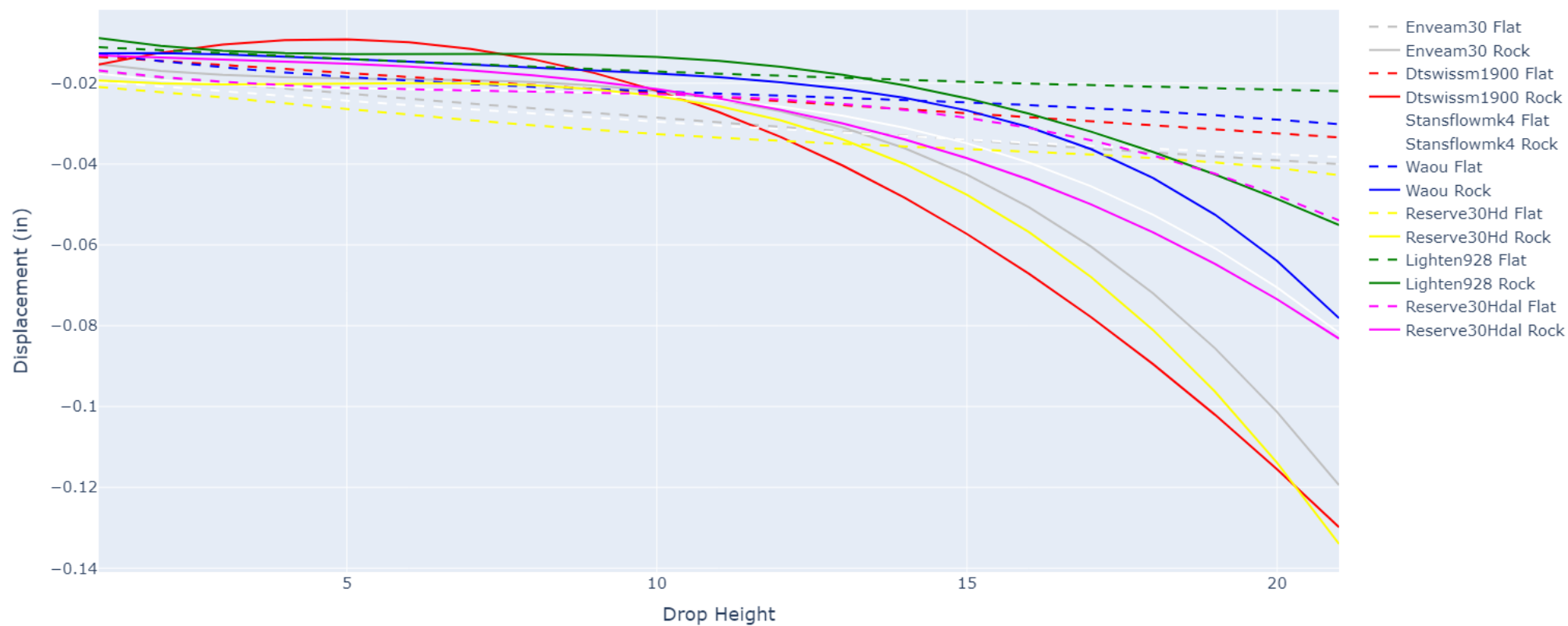
Radial Dynamic Load Test - Results

Energy Absorbed vs. Drop Heights Regression Lines, n=62



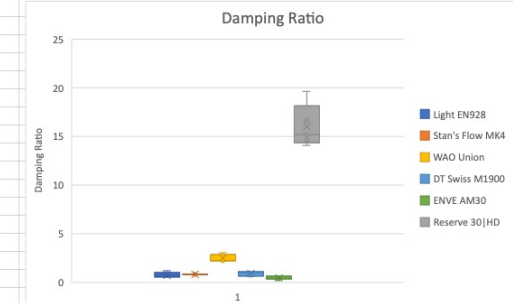
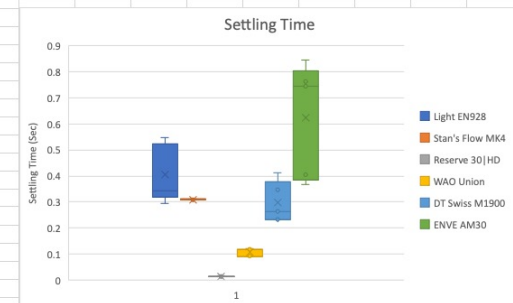
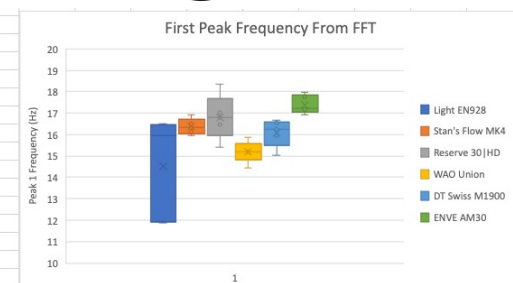
Radial Dynamic Load Test - Results

Displacement vs. Drop Height Regression Lines, n=62



Radial Dynamic Load Test - Results

Wheel		Peak 1	Peak 2	Peak 3	Peak 4	Peak 5	Avg natural freq	Settling Time (s)	Avg difference in frequency btw	Damping Ratio
Light EN928	Trial 1	15.94867	34.58749	55.33998	66.86914	80.896288	68.3515	0.3433	16.2369035	0.730569597
	Trial 2	11.87906	34.21879	54.78552	66.13268	81.203131	67.6777	0.2947	17.33101761	1.142609302
	Trial 3	11.96866	32.28756	54.55484	65.96682	80.440553	67.0404	0.5464	17.11797235	0.61165093
	Trial 4	16.43834	32.73121	54.98844	66.18979	80.44605	68.0394	0.34	16.00192678	0.715686849
	Trial 5	16.49606	32.80882	54.98685	66.16751	80.64738	68.0266	0.5	16.03783128	0.484964435
	Average	14.54616	33.32677	54.93113	66.26519	80.726681	67.82712	0.40488	16.5451303	0.679180685
Stan's Flow MK4	Trial 1	16.13927	31.78947	50.12955	64.31255	76.783806	66.1291	0.3091	15.1611336	0.801820333
	Trial 2	16.49844	32.21124	50.28096	64.73673	102.91883	73.895	0.3102	21.60509886	0.781583422
	Trial 3	16.31419	32.06582	49.97392	64.69419	103.41695	73.8023	0.3102	21.77569078	0.790410556
	Trial 4	15.95472	31.52956	50.1434	64.00881	102.75597	73.8041	0.3113	21.70031447	0.805363172
	Trial 5	16.91576	31.41498	49.94176	64.03823	104.31385	73.8482	0.3079	21.84952212	0.767995741
	Average	16.36447	31.80221	50.09392	64.3581	98.037883	72.29574	0.30974	20.41835196	0.789151929
Reserve 30 HD	Trial 1	15.41519	29.87686	72.78513	83.59165	112.515	82.8879	0.0132	24.27495176	19.65790216
	Trial 2	17.02671	31.18576	75.81365	120.0831	133.88368	99.523	0.0155	29.21424332	15.15645555
	Trial 3	18.34356	33.67763	49.72825	66.49541	81.972793	70.0986	0.0155	15.90730781	14.06839707
	Trial 4	16.47678	31.39494	77.26272	120.0133	138.49398	84.0226	0.0166	30.50430081	14.6244545
	Trial 5	16.81796	31.46587	61.57545	77.8509	121.79461	85.7949	0.0143	26.24416168	16.63223197
	Average	16.81604	31.52021	67.43304	93.60686	117.73201	84.4654	0.01502	25.22899308	15.83676
WAO Union	Trial 1	14.45528	30.13213	37.66517	58.63551	68.815281	59.5953	0.0916	13.59	3.020911359
	Trial 2	15.88126	33.29941	41.24003	63.26887	75.307888	64.4032	0.0927	14.85665818	2.717036232
	Trial 3	15.24027	32.30028	40.94401	63.69069	74.836555	65.283	0.1203	14.89907105	2.181733349
	Trial 4	15.19275	32.50541	40.72009	64.30418	75.080433	66.8153	0.1192	14.97192161	2.208754498
	Trial 5	15.18282	32.57404	41.13163	64.04388	75.085923	64.0833	0.1192	14.97577697	2.210199241
	Average	15.19047	32.16225	40.34019	62.78862	73.825216	64.03602	0.1086	14.65868556	2.42470465
DT Swiss M1900	Trial 1	16.52808	31.53385	49.80173	63.06769	76.116179	64.6211	0.234	14.89702352	1.034240665
	Trial 2	15.03503	29.51321	46.49723	58.748	70.998771	57.3559	0.2649	13.99093423	1.004323498
	Trial 3	16.68745	31.57778	49.80561	63.15557	76.248796	64.5859	0.3477	14.89033721	0.689390675
	Trial 4	16.2324	31.53724	48.9291	63.30637	75.828513	64.3996	0.4106	14.89902739	0.600147802
	Trial 5	15.94647	31.89295	50.78843	63.45823	77.111031	64.8313	0.234	15.29113924	1.071962199
	Average	16.08589	31.21101	49.16442	62.34717	75.260658	63.15876	0.29824	14.79369232	0.833775343
ENVE AM30	Trial 1	17.98774	35.97547	60.31182	68.51212	83.590073	73.8026	0.7627	16.40058394	0.291561187
	Trial 2	17.17171	36.39378	60.22914	84.32079	118.92051	87.1034	0.8433	25.43719943	0.276225844
	Trial 3	17.75249	37.37367	52.94603	60.42076	84.713647	72.0778	0.3675	16.74028876	0.61311694
	Trial 4	17.1978	36.63879	60.31692	84.24429	120.63384	84.2087	0.7439	25.85900963	0.31266017
	Trial 5	16.91134	36.55833	60.18446	83.8106	120.36893	84.5212	0.404	25.86439747	0.585464663
	Average	17.40422	36.58801	58.79767	76.26171	105.6454	80.34274	0.62428	22.06029585	0.368151115



- The vibration testing proved some results but was not easily repeatable on the same wheel or between wheels.
- Large errors and a lot of data cleaning was needed. This will need to be pursued further to find better results.

Radial Dynamic Load Test - Next Steps

- Isolate wheel mount from the rest of the stand in order to reduce vibrations travelling across the system
- Creating more impact heads to simulate a wider variety of trail scenarios
- Incorporate an improved force load sensor that can more accurately measure force input
- Incorporate new sensors or systems to measure vibrations to calculate damping characteristics with the goal of increasing repeatability and accuracy

Lateral Quasi-Static Test

Lateral Quasi-Static Test

- Goals
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Lateral Quasi-Static Test - Goals

- Determine stress thresholds in mountain bike wheelsets
- Using the MTS and OptiTrack systems together will result in the collection of data relevant to lateral stiffness.
- Simulate a low strain rate scenarios such as flat corners, aggressive berm riding, off-camber sections of trail and other uneven terrain.
 - This will equate most directly to rider weight and flow trail environment

Lateral Quasi-Static Test – Issues and Outcomes

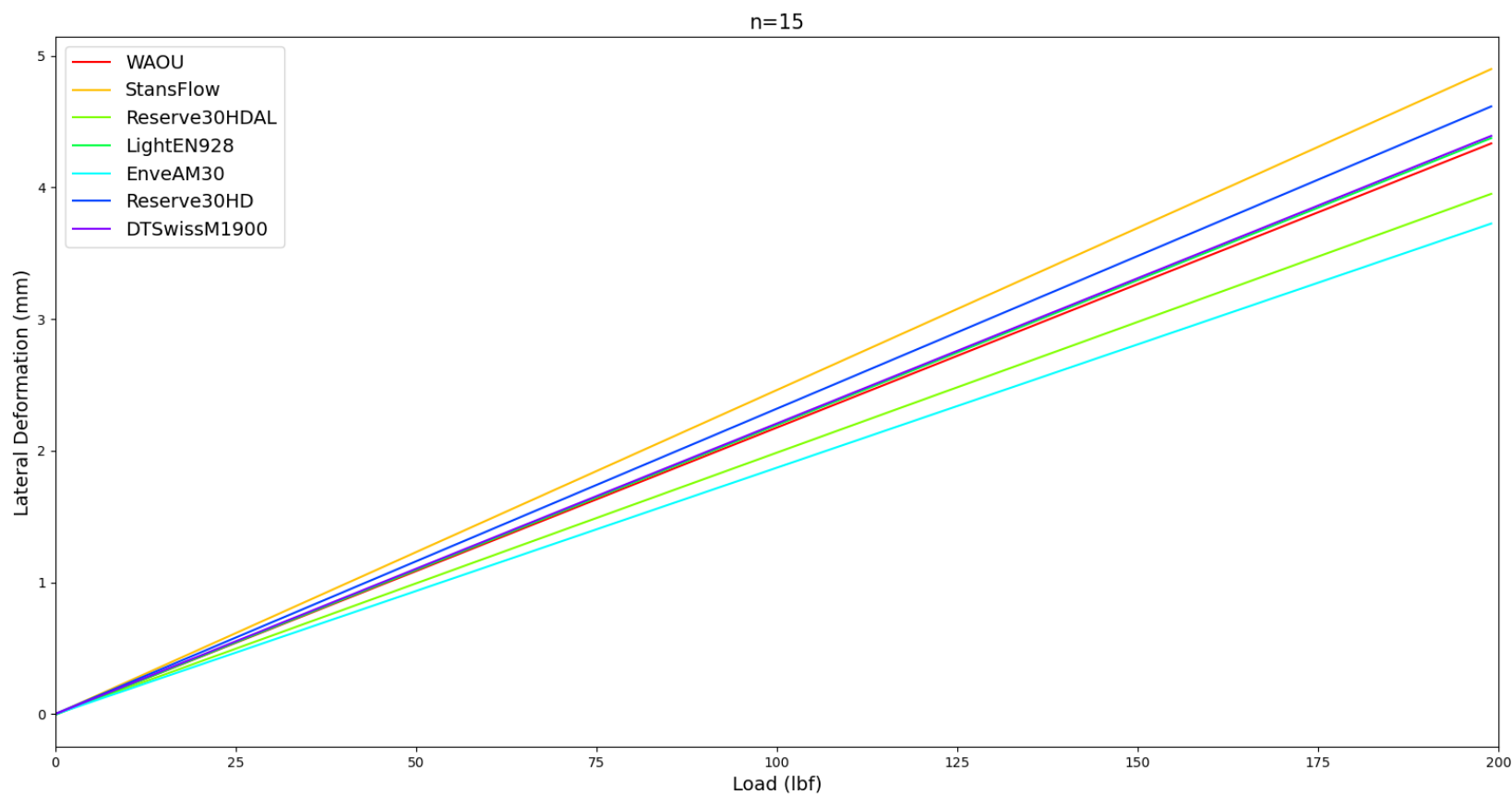
- With the current "Claw" system it is near impossible to ensure that the test is 100% repeatable for 15 trials and from wheel to wheel because of slightly different placement of the claw on the wheel
- The lateral fixture stand is slightly warped from manufacturing and can be difficult to bolt into the MTS.
- The Mocap system has a hard time tracking the markers throughout the entire test due to the stand and "Claw-System" so if there are spikes in the tracking data it means a camera either lost or found the marker which means that those cameras need to be turned off or adjusted and then recalibrated.

Lateral Quasi-Static Test – Testing Procedure

- Follow Standard Operating Procedure
 - [Standard Operating Procedure](#)
- Wheel is trued to manufacturer spec prior to each round of testing
- Ensure that tires are inflated to 32.0 psi +/- 0.1 prior to testing
- Ensure that flat circular head is removed from the MTS
- Export data as a .csv file in order to plot results

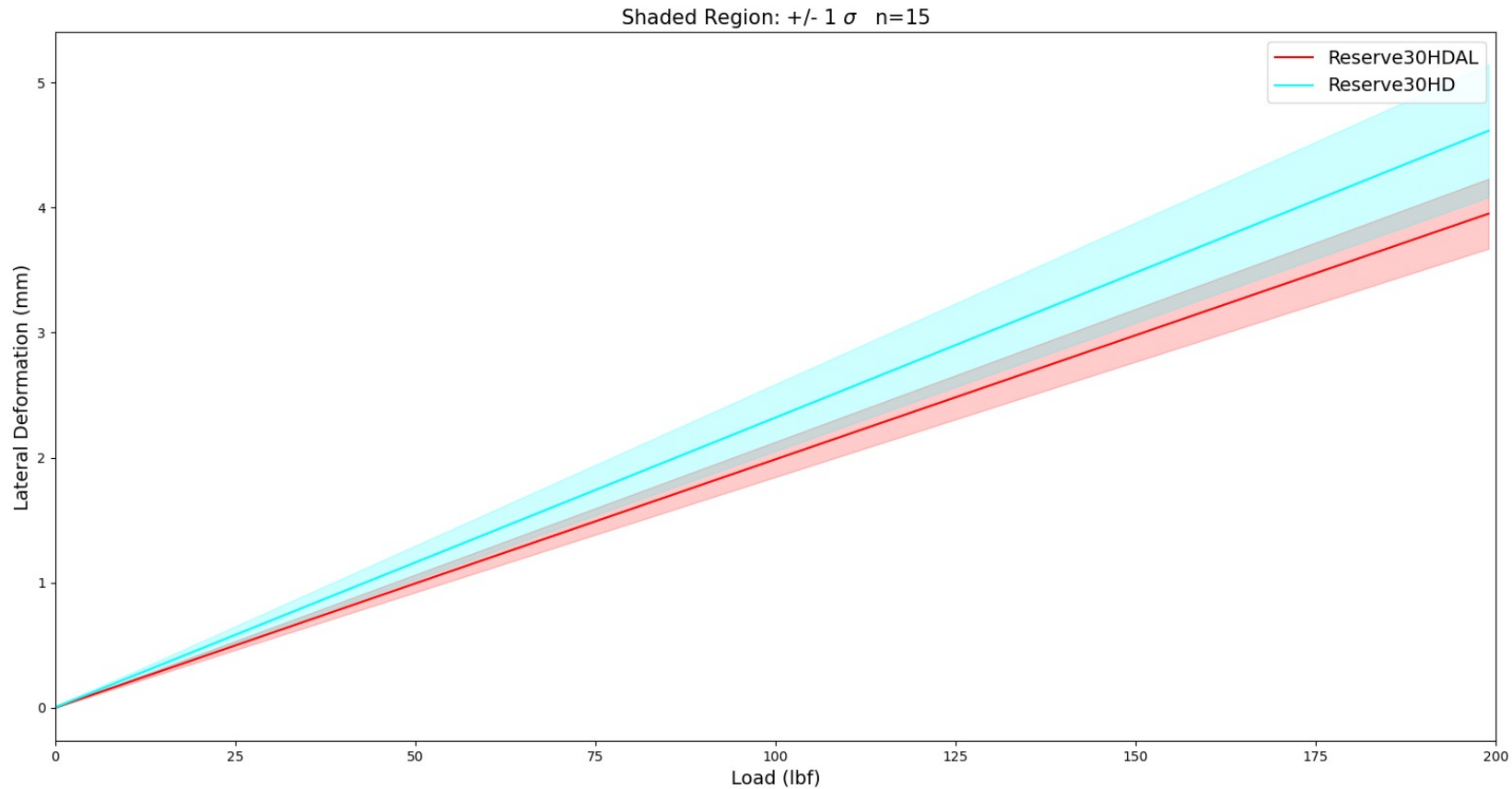
Lateral Quasi-Static Test – Results

Quasi-Static Loading on Bike Rims with Tires



Comparing Reserve HD Carbon & Aluminum

Quasi-Static Loading on Bike Rims with Tires



Lateral Quasi-Static Test – Next Steps

- The current "Claw" is slowly crumbling so a new version needs to be printed stronger or designed and manufactured using the CNC with design considerations towards accurate repeatability between trials and wheels.
 - Incorporate a way to locate the claw on the wheel and tire to increase repeatability
 - Maybe a hard stop that contacts the rim with 3 separate claw prints for the different angles

Lateral Dynamic Load Test

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Lateral Dynamic Load Test Goals

- This testing is in efforts to better understand the wheel characteristics
- Understanding how the wheel systems' harmonics, energy dissipation, and deflection are affected under different loading scenarios in hopes of matching riders with their "perfect" wheel system.

Lateral Dynamic Load Test - Issues and Outcomes

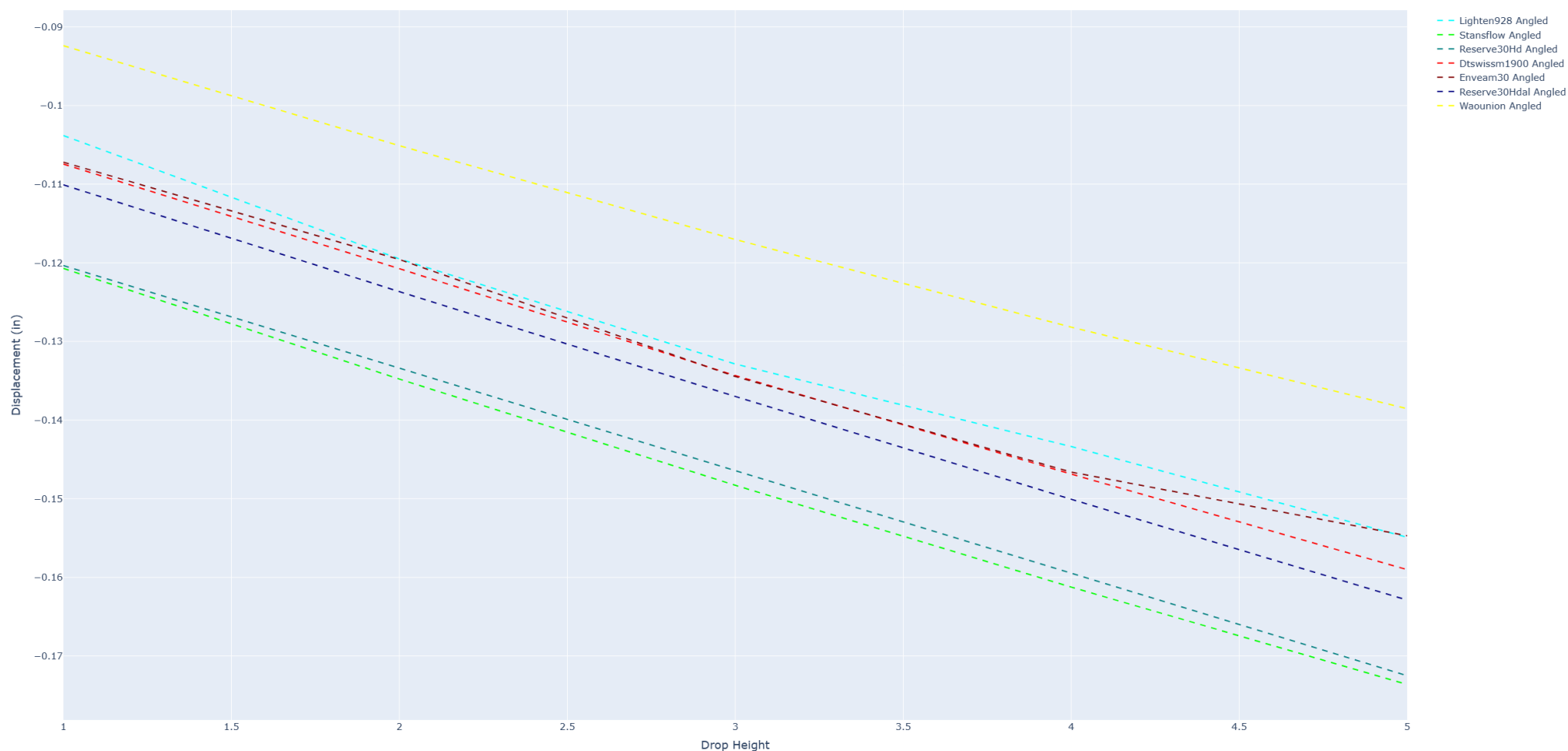
- Limited time to test slightly higher heights (**don't exceed height 7-8**, already experiencing deformation magnitudes similar to the maximum drop height on the radial tests)
- Limited time to test other angles
- No time to relate/incorporate lateral characteristics into the questionnaire/wheel comparison

Lateral Dynamic Load Test – Testing Procedure

- Follow SOP
 - [Standard Operating Procedure](#)
- Wheel is trued to manufacturer spec prior to each round of testing
- Ensure that tires are inflated to 32.0 psi +/- 0.1 prior to testing
- Install angled head to drop test apparatus
 - BE CAREFUL with the angle of the head; the maximum angle is impressively destructive; start small
- Export data as a .csv file in order to plot results

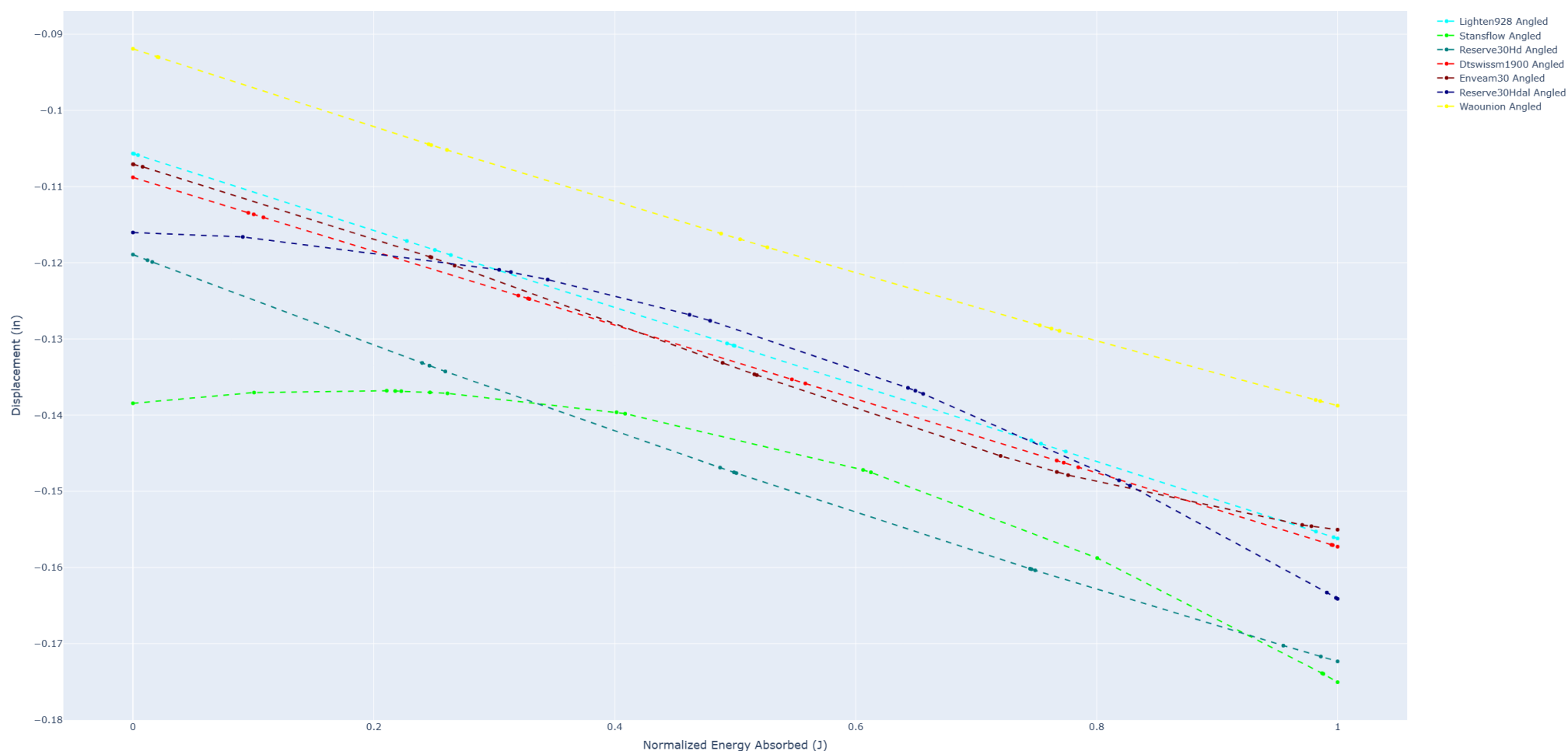
Lateral Dynamic Load Test - Results

Displacement vs. Drop Height Regression Lines, n=62



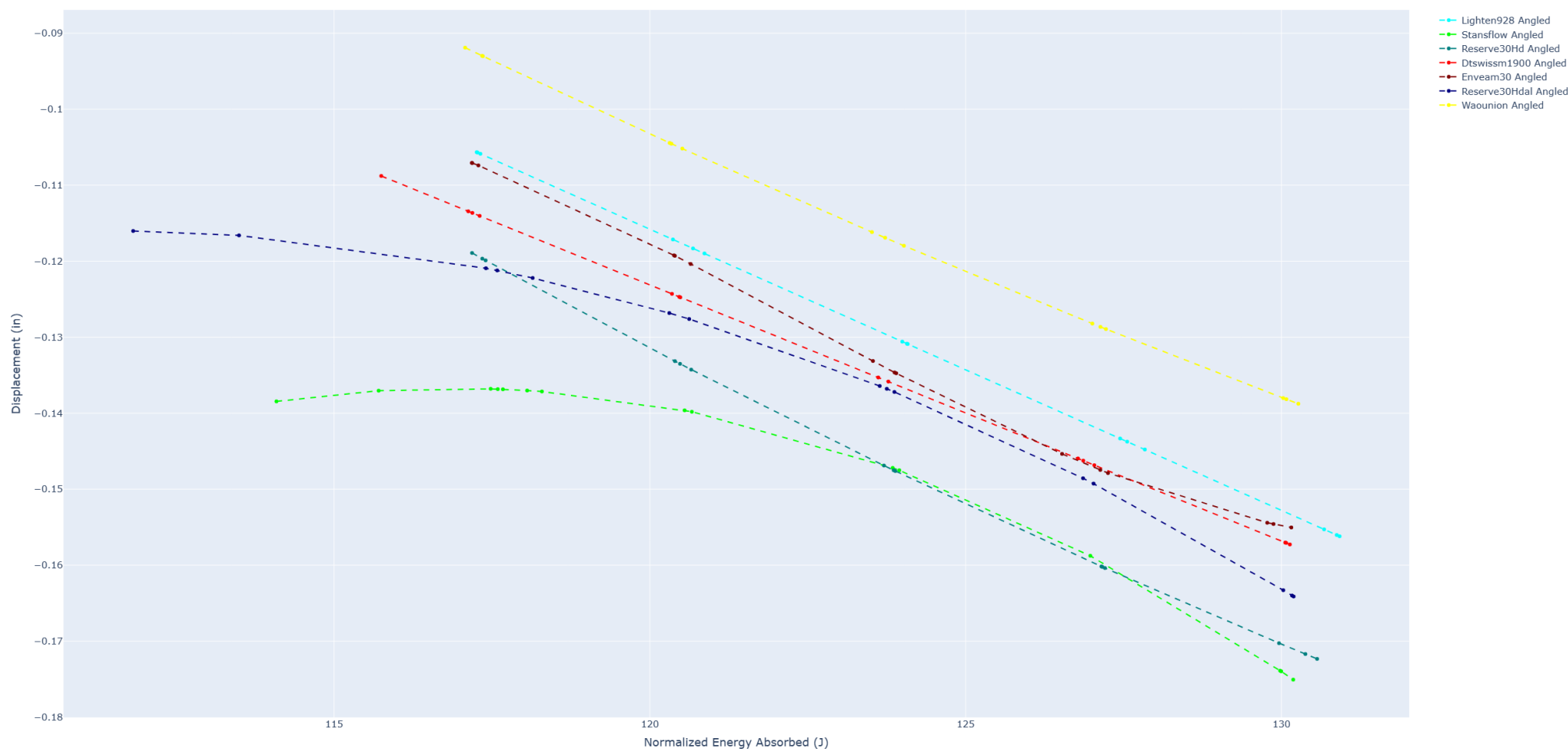
Lateral Dynamic Load Test - Results

Energy Absorbed vs. Displacement Regression Lines, n=62

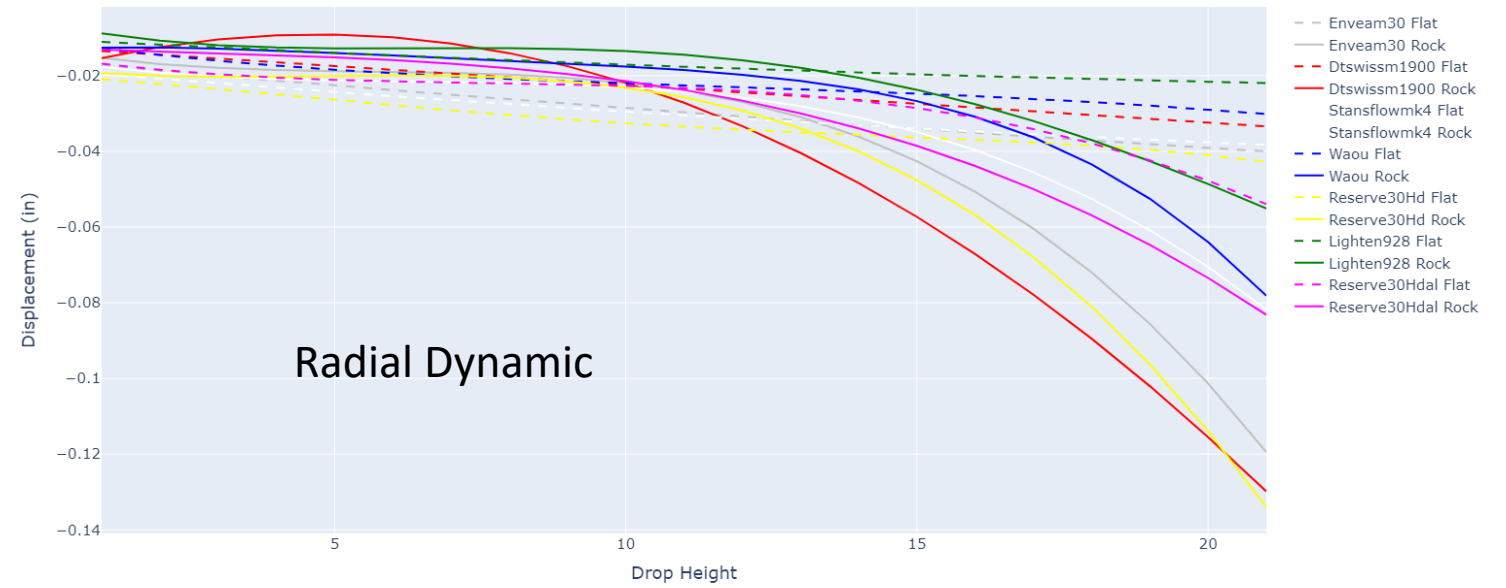


Lateral Dynamic Load Test - Results

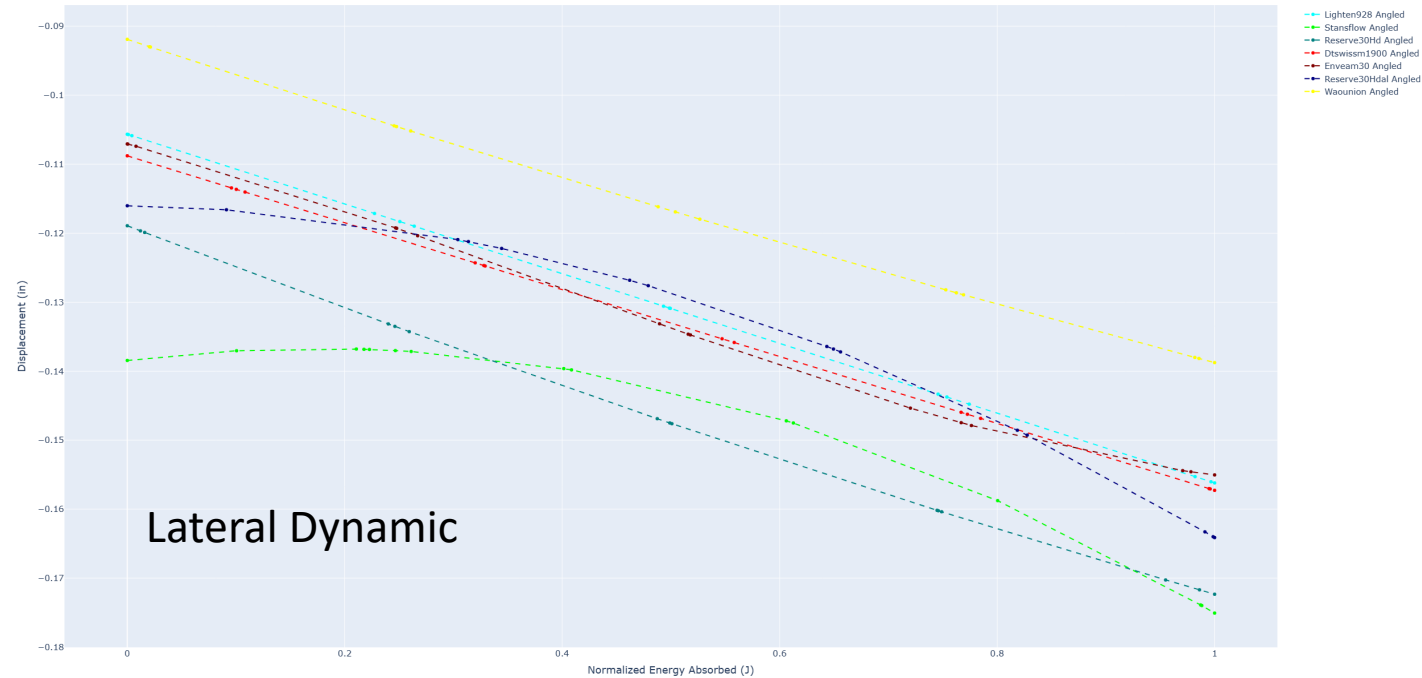
Energy Absorbed vs. Displacement Regression Lines, n=62



Displacement vs. Drop Height Regression Lines, n=62



Energy Absorbed vs. Displacement Regression Lines, n=62



Lateral Dynamic Load Test - Next Steps

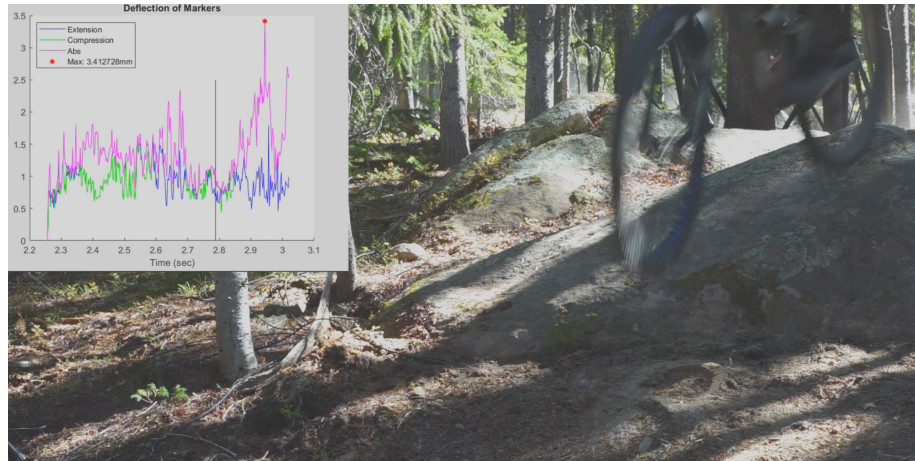
- Measuring natural frequency and settling time to determine damping coefficients to provide more information on a wheel's ability to absorb energy in the lateral direction. (Hopefully validates energy dissipation)
- Test more angles on impact head
- See if there are any deviations in trend at slightly higher drop heights
- Incorporate lateral characteristics into the wheel categorization and rider wheel questionnaire

Field Data

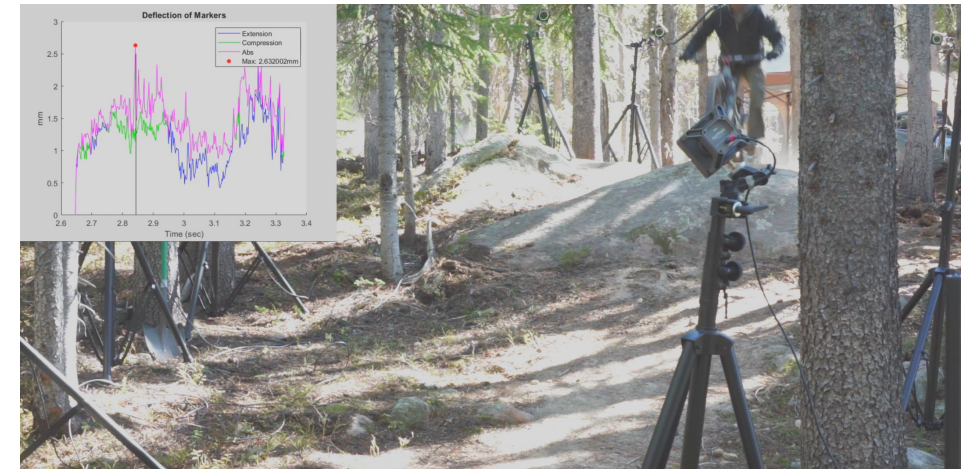
Field Data – Goals

- Quantitatively and qualitatively verify in lab deflection and wheel performance data through capturing and analyzing a variety of real-world riding scenarios using the OptiTrack system to track markers placed on the rim of a wheel
- Use the field data to inform in lab testing and produce comparable results to compare wheels in a controlled environment

Field Data – Summer 2022



Stans Flow off test track drop
experienced 3.4mm of radial deflection



Light EN928 off test track drop
experienced 2.6mm of radial deflection

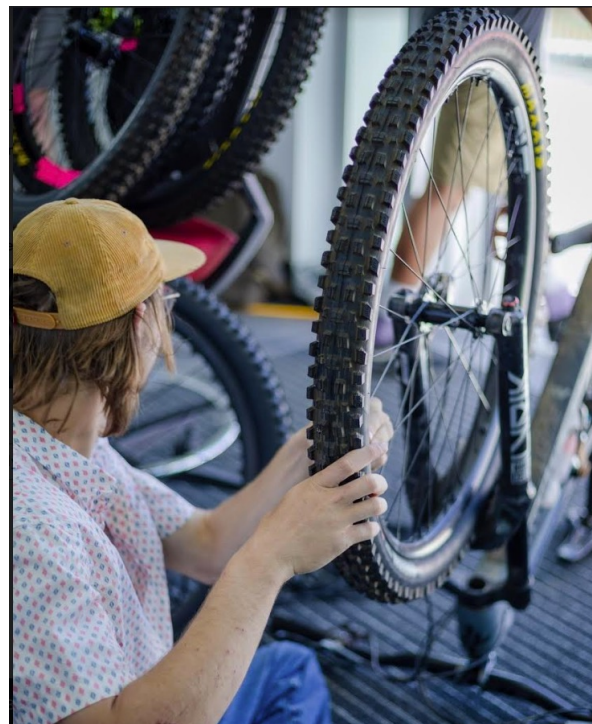
Field Data – Summer 2022 Issues and Outcomes

- Filtering process for useful markers is very aggressive and removes more than half of the markers in some scenarios and in other scenarios it makes a set of data unusable (had to be done due to the extremely sporadic data sets, solution is to collect more consistent data), analysis method is good though
- Data sets themselves are extremely variable (only 3-4 useable sets of data), no rigid bodies were implemented so it makes it hard to know which marker is which (allows you to label each marker within a rigid body), and we discovered that OptiTrack is optimized to track rigid bodies
- Within the usable sets of data there are weird instances of marker "float" where markers will move over 4mm unexpectedly
 - we assume a combination of calibration quality (needs to be less than 0.25mm at worst but "excellent" in OptiTrack can be up to 0.5mm :/), lack of rigid bodies, and lighting caused this movement



Field Data – Summer 2023

- Goals were to test the use of rigid bodies on the wheel and hub, improve calibration accuracy, and more controlled lighting
- We tested on both a small drop and staircase here at Rady
- Brief sections of data look better than the previous collection, but due to the lack of cameras there isn't a long enough section of consistent data to get relevant measurements, we recommend another round of data collection using the recommended improvements on the next slide



Field Data – Summer 2023 Issues and Outcomes

- Found that the use of rigid bodies on the hub and wheel is the way to go because it results in better marker tracking, easier sorting, and additional rigid body rotation and position data (the use of a rigid body does not affect the individual marker positions so there is no reason not to use rigid bodies)
- Main issue being the use of only 6 cameras resulting in inconsistent and sporadic data sets, making it impossible to pull anything meaningful out of them (recommend at least 8+ cameras), it also resulted in too small of a capture area preventing the system's ability to pick up the points of interest
- Would recommend the use of a larger obstacle to get a more obvious point of deformation, as we found that even with the best calibration, we still encountered residuals greater than 0.4mm which is around 25% of the expected deformation
- Again, inconsistent lighting causes the need to constantly recalibrate and sporadic data collection (sun bad)(recommend trying to test in the Field House on campus to nail down a testing procedure)
- Kept losing markers and picking them back up again as they passed the fork leg



Wheel Questionnaire

Wheel Questionnaire - Goals

- Create a deliverable questionnaire accessible to consumers (all skills of rider) that calculates the perfect wheel for the rider
- Calculates the best wheel for the rider by finding values for the rider in different categories that we can correlate to lab data for the wheels

Wheel Questionnaire - Details

	A	B	D	E	F	G	H	I	J	K	L	M
1	Rider Criteria	Enter Responses Below										
2												
3	Rider Weight:	165										
4	<i>in lbs</i>											
5	Rider Aggression:	9										
6	<i>Rank 1-10: 1 = Very conservative (not okay with falling ever), 5 = Interested in expanding comfort zone/abilities, 10 = Very aggressive (always out of your comfort zone)</i>											
7	Rider Finesse-Power:	8										
8	<i>Rank 1-10: 1 = I feel light and precise when I ride; always looking for the smoothest line choice, 5 = I can be precise when I need to be but I'm okay with smashing, 10 = I'm a monster truck</i>											
9	Trail Environment:	9										
10	<i>Rank 1-10: 1 = Smooth cross-country style trail, 5 = Trail with some technical features, 10 = Big rocks / sharp edges / big gap jumps</i>											
11	Front Suspension Travel:	170										
12	Rear Suspension Travel:	170										
13	<i>in mm</i>											
14												

Wheel Questionnaire - Details

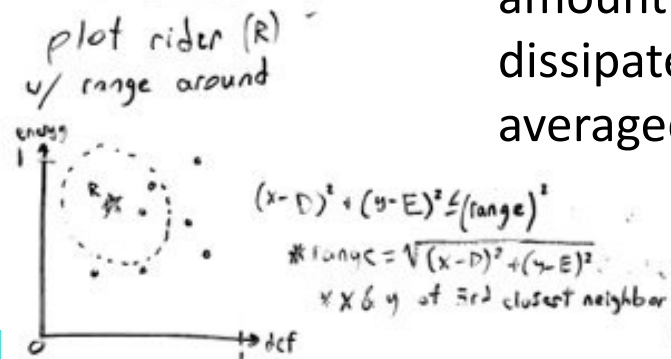
- The defining qualities we selected to match wheels with riders were energy dissipated by the wheel and deformation of the wheel
- We ran an excel regression to calculate equations to produce each normalized value
- We used five independent variables (questions); rider weight, aggression, finesse vs power, trail environment, and suspension travel

Regression Equations @ weight
 when all scaled 1-10; @ = aggression
 (after scaling min/max) @ = finesse @ = environment
 ④ = avg travel

$$\textcircled{D} = 0.993 \left[(0.0533) \cdot w + (0.0326) \cdot a + (0.0205) [0.08 \cdot f \cdot e + 0.8] + (0.0080) \cdot t - 0.112 \right]$$

$$\textcircled{E} = 0.9465 \left[(0.0176) \cdot w + (0.0473) \cdot a + (0.0329) [0.08 \cdot f \cdot e + 0.8] + (0.0235) \cdot t - 0.117 \right]$$

- In order to produce an accurate equation, two of the experienced riders in our group estimated the amount of deformation/compliance and energy dissipated that each rider in our sample needed and averaged our results for our dependent variables



Equations for the normalized values
 of energy dissipated and deformation

$$=0.9465*((0.0176*J21)+(0.0473*K21)+0.0329*(0.08*L21*M21+0.8)+(0.0235*N21)-0.117)$$

G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Data from Questionnaire			lbs	out of 10	out of 10	out of 5	mm	Normalized Force Values				Josh Calculation		Zach Calculation			Regression Equation Calculation!		
		Rider	Weight	Aggression	Finesse	Environme	Avg Travel		Deformati	Energy		Deformati	Energy	Deformati	Energy		Deformati	Energy	
		Zach	165	9	8	4.5	170		0.775	0.825		0.800	0.900	0.650	0.750		0.695	0.774	
		Josh	160	8	7	4.5	170		0.670	0.710		0.700	0.700	0.640	0.710		0.595	0.691	
		Tyler	145	8	4	4.0	155		0.550	0.675		0.600	0.700	0.500	0.650		0.536	0.591	
		Skylar	205	7	8	4.5	155		0.800	0.705		0.900	0.700	0.690	0.710		0.728	0.696	
		Enduro Cai	115	6	5	4.0	155		0.450	0.415		0.500	0.400	0.350	0.430		0.379	0.488	
		XC Cam	115	6	4	1.0	100		0.350	0.275		0.400	0.200	0.350	0.350		0.295	0.319	
		Mel	160	3	2	2.0	110		0.375	0.215		0.300	0.100	0.450	0.330		0.308	0.236	
		Jacob	190	7	8	3.5	160		0.735	0.750		0.800	0.800	0.670	0.700		0.657	0.662	
		Charlie	195	7	4	3.0	160		0.685	0.550		0.700	0.500	0.670	0.600		0.658	0.599	
		Luke	155	8	7	4.5	170		0.670	0.710		0.700	0.700	0.640	0.720		0.595	0.691	
		Erik	190	6	7	2	0		0.600	0.400		0.600	0.500	0.600	0.400		0.523	0.391	
		Emily	115	7	5	3	154		0.440	0.520		0.500	0.600	0.360	0.460		0.395	0.508	
		Ethan	175	6	6	5	155		0.650	0.600		0.700	0.600	0.600	0.600		0.521	0.513	
		Maxwell	240	10	10	5	200		1.000	1.000							1.000	1.000	
		Min-Lee	80	1	1	1	60		0.000	0.000							0.000	0.000	
		Data Normalized to 1-10 Scaling							Environment scales Finesse										
		Rider	Weight	Aggression	Finesse	Environme	Avg Travel		Rider	Weight	Aggression	Fin/Env	Avg Travel						
		Zach	6.000	9.000	8.000	9.000	8.000		Zach	6.000	9.000	7.452	8.000						
		Josh	5.000	8.000	7.000	9.000	8.000		Josh	5.000	8.000	6.634	8.000						
		Tyler	5.000	8.000	4.000	8.000	7.000		Tyler	5.000	8.000	3.817	7.000						
		Skylar	8.000	7.000	8.000	9.000	7.000		Skylar	8.000	7.000	7.452	7.000						
		Enduro Cai	3.000	6.000	5.000	8.000	7.000		Enduro Cai	3.000	6.000	4.544	7.000						
		XC Cam	3.000	6.000	4.000	2.000	3.000		XC Cam	3.000	6.000	1.636	3.000						
		Mel	5.000	3.000	2.000	3.000	4.000		Mel	5.000	3.000	1.454	4.000						
		Jacob	7.000	7.000	8.000	7.000	8.000		Jacob	7.000	7.000	5.998	8.000						
		Charlie	8.000	7.000	4.000	6.000	8.000		Charlie	8.000	7.000	3.090	8.000						
		Luke	5.000	8.000	7.000	9.000	8.000		Luke	5.000	8.000	6.634	8.000						
		Erik	7.000	6.000	7.000	4.000	1.000		Erik	7.000	6.000	3.453	1.000						
		Emily	3.000	7.000	5.000	6.000	7.000		Emily	3.000	7.000	3.635	7.000						
		Ethan	6.000	6.000	6.000	5.000	7.000		Ethan	6.000	6.000	3.635	7.000						
		Maxwell	10.000	10.000	10.000	10.000	10.000		Maxwell	10.000	10.000	9.997	10.000						
		Min-Lee	1.000	1.000	1.000	1.000	1.000		Min-Lee	1.000	1.000	1.000	1.000						

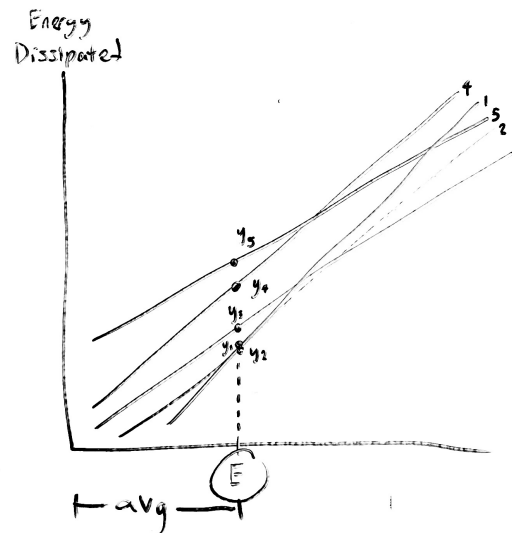
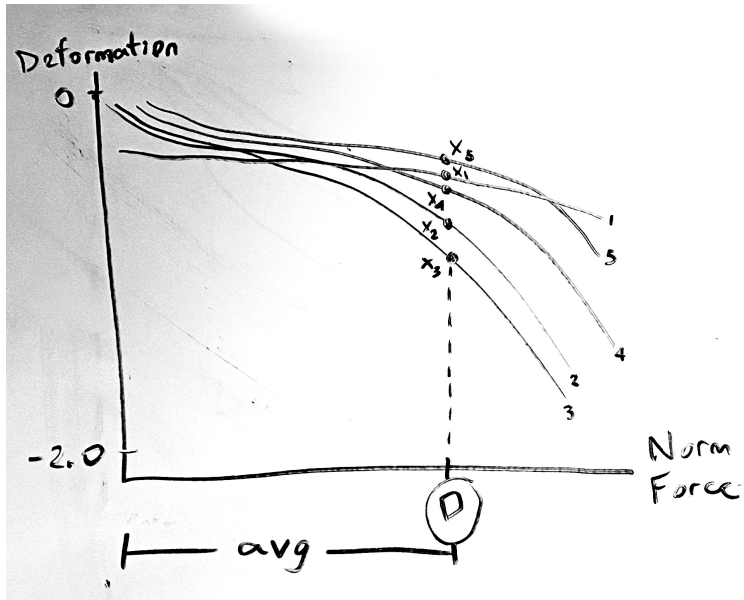
Wheel Questionnaire.xlsx can
be found in the Blister Labs
Summer Wheelsets OneDrive

Wheel Questionnaire - Issues and Outcomes

- We established equations to produce normalized values for energy and deformation based on rider input using a large sample including control riders to establish minimum and maximum input values
- We could not correlate our rider values to wheels since we only had all mountain/enduro wheels on test; without an XC and DH wheelset we cannot normalize our energy and deformation values for the wheels

Wheel Questionnaire - Next Steps

- Correlating rider data to wheel data once good field data is acquired
- Acquiring a larger sample of wheels in order to normalize values for every type of rider (Create a database to use as reference for normalization)
 - Once the wheel values are normalized from minimum to maximum, we can plot the wheels using the average value up to the deformation/energy value found in our deformation/energy vs normalized force plots



- Once plotted on a Deformation vs Energy plot, we can find the 3 closest neighbors to each riders' normalized values

